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Cognitive appraisals, distress and disability among persons in low vision rehabilitation

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Objective. Examined cognitive appraisals of interference and tolerance in the prediction of distress and self-reported disability among persons presenting for low vision rehabilitation.

Design. Cross-sectional; correlational and path analyses.

Methods. One-hundred and thirteen patients (mean age, 71 years; 52 men and 61 women) presenting for low vision rehabilitation at a university-based centre for low vision rehabilitation participated in an initial clinical vision examination and completed several questionnaires to evaluate cognitive appraisals, emotional distress and self-reported disability.

Results. Path analyses indicated that greater tolerance was associated with less interference imposed by vision loss. Greater tolerance was also associated with less emotional distress and symptom severity (visual acuity) was associated with self-reported disability. Cognitive appraisals (tolerance and interference) indirectly influenced self-reported disability through emotional distress.

Conclusions. The data indicate that appraisals of personal ability to tolerate vision loss and the perceived interference of vision loss on goal-directed behaviour and expected activities have greater influence on distress and are subsequently predictive of disability in comparison with objective symptoms (visual acuity). Implications for clinical interventions and further research are discussed.

Accumulating evidence indicates that there is considerable variation in the ways persons with visual impairments react and adapt in the wake of vision loss (e.g. Brennan,

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2002; Dreer, Elliott, Fletcher, & Swanson, 2005; Schilling & Wahl, 2006; Teitelbaum & Copolillo, 2005). For example, it has been argued that a major loss does not necessarily lead to depression or distress for some (Kleinschmidt, 1999) and yet for others, persistent depression may continue for longer than normally expected (Ciechanowski, Katon, & Russo, 2000). Additional support for the influential role of intra-personal factors underlying depression associated with vision impairment has also been demonstrated (Tolman, Hill, Kleinschmidt, & Gregg, 2005). In an effort to find explanations for the wide variations in reactions related to the adjustment to disability, contemporary models of adjustment and coping have been informative (Dodds *et al.*, 1994; Hayeems, Gellar, Finkelstein, & Faden, 2005).

One of the most widely accepted models, the transactional model of stress appraisal and coping, posits that cognitive *appraisals* of a stressor and *coping strategies* are important factors that can have an influence on adaptation outcomes (Lazarus & DeLongis, 1983). Individuals presumably appraise a stressful situation for threats to their well-being (*primary appraisal*) and to assess if their behaviour can have an impact on the stressor (*secondary appraisal*). Thus, the central feature of this model is the subjective, cognitive appraisal of the situation-specific event (rather than an objectively defined stressor, i.e. severity of vision loss) and one's coping ability; both of which are considered critical to successfully managing a stressful situation (Folkman & Lazarus, 1985; Oliver & Brough, 2002). Thus, from this viewpoint, stress is the culmination of the dynamic interplay between the individual and the environment.

Converging evidence indicates that phenomenological appraisals of stress, personal resources and subjective meaning are predictive of personal adjustment following chronic disease and disability (Elliott & Harkins, 1991, 1992; Elliott, Kurylo, & Rivera, 2002; Harkins, Elliott, & Wan, 2006). Generally, this work has examined appraisal processes relevant to various health conditions that yield considerable information beyond the concepts of primary and secondary appraisals (Ptacek & Pierce, 2003). Chronic health conditions can disrupt self-regulatory behaviour and outcome expectancies, and these disruptions can trigger a variety of affective responses (Orbell, Hagger, Val Brown, & Tidy, 2004; Williamson, 1998). These key findings are consistent with the investigations into goal-directed routine behaviour that is interrupted or non-rewarded for overlearned responses (Averill, 1982, 1983; Berkowitz, 1989) based on the notion that individuals utilize cognitive schema comprising expected behaviours, goals, rewards and reinforcers in the ongoing interactions with the environment. Therefore, when events occur that are incongruent with these expectations, a person will automatically appraise the degree to which the event interferes with goal-directed behaviours and previously anticipated outcomes.

Individuals appraise the extent to which specific health conditions may interfere with their expected behaviours and goal-directed pursuits. They then appraise the degree to which they may be able to *tolerate* disruption to expected behaviours and goal-directed pursuits (Elliott & Harkins, 1992; Harkins *et al.*, 2006). Individuals differ, however, in their perceived ability to tolerate disruptions to the expected behaviours and goal-directed pursuits imposed by the stress between their personal needs and environmental reinforcers (Dawis & Lofquist, 1984). This may involve efforts to exert control over internal emotional states upon realizing that behavioural control over specific events or symptoms may not be possible (Heckenhausen & Schulz, 1995).

Cognitive appraisals have substantial effects on emotional distress associated with a chronic disease, and this association can be stronger than the effects on distress attributable to disease symptom severity (Harkins *et al.*, 2006). However, we do not

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know the degree to which these cognitive appraisals predict functional disability, as assessed on self-reported measures frequently used in clinical research protocols. It is possible that these reports may be directly influenced by the degree to which sight loss interferes with goal-directed pursuits (i.e. interference appraisals), as one's appraisals may be based, in part, on the disruptions in daily activities that an individual experiences from the condition. We do not know if the relation of interference to the vision-related self-reported disability would remain significant once objective indicators of symptom severity are taken into account. It is also possible that the degree to which an individual can tolerate these disruptions may contribute to the prediction of vision-related disability, independent of the variance attributable to perceived interference (e.g. Harkins *et al.*, 2006).

The application of these concepts related to the transactional model of adjustment has recently gained increasing attention in the vision literature (Boerner, Reinhardt, & Horowitz, 2005; Jackson & Taylor, 2000; Ringering & Amaral, 2000). However, in comparison with the studies examining coping responses (i.e. Boerner, 2004; Brennan *et al.*, 2001; Ryan, Anas, Beamer, & Bajorek, 2003), only a few empirical investigations of a person's appraisal of the condition of vision loss or its concomitants have been documented in the vision literature. For instance, Rovner and Casten (2002) found that the relationship between visual acuity and depression is mediated by the loss of valued, discretionary activities. From our perspective, personal reactions are directly determined by the degree to which a chronic health condition interferes with desired pursuits, and with the degree to which an individual can tolerate these disruptions (Harkins *et al.*, 2006).

In the current study, we were interested in further examining the relation of the appraisal processes and an objective indicator of symptom severity (i.e. visual acuity) to both the vision-related self-reported disability and the emotional distress reported by persons presenting to low vision rehabilitation services. A significant minority of persons with vision loss experience problems with depression and distress (Casten, Rovner, & Edmonds, 2002; Horowitz & Reinhardt, 2000; Scott, Schein, Feuer, Folstein, & Bandeen-Roche, 2001), and distress is associated with impaired activities of daily living (Heyl & Wahl, 2001; Ryan *et al.*, 2003). We argue that emotional distress associated with vision loss is directly influenced by cognitive appraisals of the loss. In contrast to other studies that have relied upon self-reported measures of the stress, we used an objective index to quantify actual dysfunction (e.g. visual acuity). We hypothesized that reports of greater interference from changes in vision on expected behaviour would be significantly predictive of greater emotional distress and vision-related self-reported disability, above and beyond objective indicators of vision impairment (e.g. visual acuity). We predicted that distress associated with vision loss would be determined more by subjective, cognitive appraisals than by objective visual impairment severity. Based on previous research (Harkins *et al.*, 2006), we expected that greater tolerance of loss of vision would contribute to the prediction of lower distress; however, we were uncertain if tolerance to loss of vision would be associated with self-reported disability.

Methods

Participants

Participants completed information as part of their initial clinical evaluation in a university-based low vision rehabilitation clinic. The sample included 52 men

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(*M* age = 71.23, *SD* = 14.93) and 61 women (*M* age = 70.49, *SD* = 18.13) who participated in a low vision evaluation upon admission to an out-patient low vision rehabilitation programme. Participants' better eye distance visual acuity logMAR scores ranged from .0 to 1.6 (*SD* = 0.82) and eight participants' visual acuity was documented as 'hand motion only' or 'no light perception'. The sample averaged 13 years of formal education (*SD* = 2.92). Eight percentage of the participants were employed (*N* = 9); 71.5% were retired (*N* = 80); 11.5% were disabled (*N* = 13); 3.5% were unemployed (*N* = 4); and 6.2% were missing data (*N* = 7). Twenty-seven percentage were widowed (*N* = 31); 5% were divorced (*N* = 6); 50% were married (*N* = 56); 1% was separated (*N* = 1); 16% were single (*N* = 18); and 1% was missing data (*N* = 1). Macular degeneration was the most frequent primary referring diagnosis for the sample (57.5%, *N* = 65); approximately 4.4% (*N* = 5) were diagnosed with diabetic retinopathy as a primary diagnosis; 4.4% (*N* = 5) were diagnosed with glaucoma; 2.7% (*N* = 3) were diagnosed with cataracts; 27.4% (*N* = 31) were diagnosed with other various vision impairments (e.g. Stargardt disease, etc.); and 3.5% (*N* = 4) were given a diagnosis of vision impairment of unknown origin upon intake. One-hundred and two persons were Caucasian and 11 persons were African-American.

A version of the original *National Eye Institute-Visual Function Questionnaire* (NEI-VFQ: Mangione *et al.*, 1998) was mailed to each participant prior to their appointment date. Although participants were instructed to complete this measure prior to their appointment, some participants arrived with incomplete questionnaires and subsequently completed the modified version of the NEI-VFQ in the clinic. The majority of participants completed the questionnaire with the assistance of a family member. Following participants' initial eye examinations, a doctoral-level clinical psychologist met individually with participants and read all the instructions and administered measures.

Measures

Cognitive appraisal variables

Two separate tactile analogue scales (TAS) (Dreer *et al.*, 2005) were employed to evaluate appraisals of interference and tolerance. Each TAS was presented in the form of a 10 ml syringe without an attached needle. The measure of tolerance was assessed by asking participants, '*How difficult is it for you to tolerate your changes in vision?*' The appraisal was anchored by the phrase '*no difficulty*' = 0 and at the other end '*greatest difficulty imaginable*' = 100, with responses reversed scored so that lower scores indicate greater difficulty tolerating vision changes. Participants rated the degree of interference with daily activities on an item that was phrased, '*How much do your changes in vision prevent or interfere with what you want to do?*' This scale was anchored at opposite ends by descriptors '*no interference*' = 0 and '*completeinterference*' = 100. Higher scores indicate perceptions of greater interference. The TASs used in this study were adapted from visual analogue scales (VAS) that have been successfully used in previous studies of interference and tolerance appraisals of pain (Elliott & Harkins, 1992; Harkins, Price, & Braith, 1989), chronic stress (Elliott, Chartrand & Harkins, 1994) and urinary incontinence (Harkins *et al.*, 2006). Although these appraisals are predictably associated with the measures of depression and trait measures of negative affectivity, they maintain significantly and contribute independently to the prediction of distress reported by community-residing adults (Elliot *et al.*, 1994; Harkins *et al.*, 2006).

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Vision symptom severity

Visual acuity was obtained by using the early treatment diabetic retinopathy study charts (ETDRS). ETDRS ratings were then converted to the logMAR scale (Bailey & Lovie, 1976). For the purposes of this study, we used the logMAR visual acuity score for the better eye. Higher scores for visual acuity indicate greater vision impairment.

Emotional distress

TASs were used to evaluate negative emotions specifically attributed to vision loss (Dreer *et al.*, 2005). These scales were concerned with feelings which might be expected to be evoked by vision loss or by thinking about the condition; these included anger, fear, depression, anxiety and frustration. Each TAS was presented in the form of a 10 ml syringe without an attached needle. Each emotion TAS was anchored by the verbal descriptors '*none at all*' = 0 and '*the most severe imaginable*' = 10. Participants were instructed to indicate the extent of depression, anger, fear, frustration and anxiety, by pulling the syringe tube between the anchors. Higher scores on each scale indicate greater negative affect. The five items were averaged to provide a single indicator of distress.

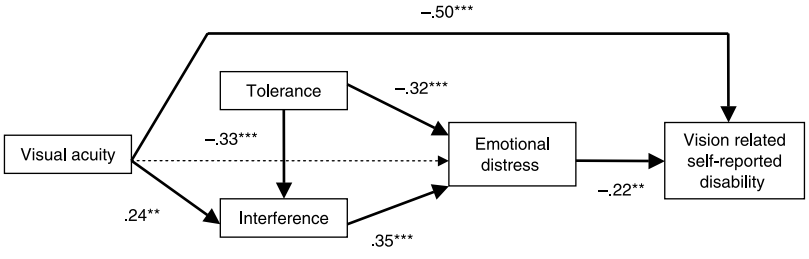
Internal consistency for the averaged index score of emotional distress associated with vision loss was acceptable ($\alpha = .77$; $N = 92$). Similar alphas have been observed for a variation of this instrument VAS in research with older clinical samples (.89) (Harkins *et al.*, 2006), undergraduate students (.70, .71) (Elliott, Sherwin, Harkins, & Marmarosh, 1995) and undergraduate women (.86) (Elliott & Harkins, 1992). Higher VAS emotional distress scores are positively and significantly related to negative affectivity (Harkins *et al.*, 1989). These studies support the validity of an emotional distress average score as a meaningful measure of domain-specific distress associated with a health problem.

To evaluate the validity of the syringe analogue measurement of these dimensions, a subsample of consecutively referred participants ($N = 13$) completed a second administration by using a large emotional 'thermometer' developed for the clinic and this study. The thermometer was made with poster board and divided into 10 equal intervals from 0 to 10 (*most severe imaginable*). All participants were able to correctly identify the figure as a thermometer with numbers. Participants were given the same instructions and asked to indicate on the thermometer the intensity of each emotion. Participants either pointed directly on the thermometer to denote their response or verbally gave a number to the examiner. The correlation between the average score from the syringe administration and the thermometer indicated a moderately high degree of reliability ($r = .88$, $p < .001$).

Vision-related self-reported disability

An abbreviated version of the NEI-VFQ (Mangione *et al.*, 1998) was used to evaluate the impact of visual impairment on daily functioning. The scale was shortened by the fourth author (DF) to contain items that were considered clinically important for the vision rehabilitation team to develop individualized rehabilitation programmes. This reduced the measure to 23 of the 51 original NEI-VFQ items (Massof & Fletcher, 2001) and 5 of the original 13 domains: near vision (7 items), distance vision (7 items), social functioning (3 items), dependency (4 items) and mental health (2 items). Responses for 17 of the items include the following range: 1 = '*no difficulty at all*', 2 = '*a little*

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Note. Lower scores on tolerance indicate greater difficultyin the ability to toleratevision-related changes; higher scoresfor disability reflect fewer problems associated with vision-specificdisability and higher scores for visual acuity indicate greater vision impairment.

Figure 1. Path diagram of the model tested to evaluate the impact of symptom severity (acuity), appraisals of tolerance and interference, and emotional distress on vision-related self-reported disability.

difficulty', 3 = '*moderate difficulty*', 4 = '*extreme difficulty*', 5 = '*stopped doing this because of your eyesight*' and 6 = '*stopped doing this for other reasons or not interested in doing this*'. The other six items require frequency or level of agreement ratings. Similar to the NEI-VFQ, this version is scored by linearly transforming a respondent's ratings for each item to values that range from 0 to 100. Scores are combined to produce a total impairment score (Parrish *et al.*, 1997), with higher scores indicating greater vision-specific disability. For this study, the sum of the 23 items was used as a total vision-related disability score. The Cronbach's α coefficient for the measure was .94 and the corrected item – total correlations ranged from .44 to .79.

Statistical analysis

Path analysis was used to evaluate the hypothesized relationships between the predictor variables (i.e. cognitive appraisals, vision symptom severity variables, emotional distress) and vision-related self-reported disability as the main outcome or endogenous variable. Path analysis permits the examination of the indirect and direct relationships among the variables. The model evaluated the specific hypotheses concerning the causal relations of vision symptom severity (i.e. visual acuity), cognitive appraisals (i.e. tolerance and interference) and emotional distress on vision-related self-reported disability (Figure 1). Path analysis permits the evaluation of how well the collected data fit the hypothesized model (Byrne, 1989). The Comparative Fit Index (CFI) (Bentler, 1990) was used to evaluate the model. The CFI ranges from 0 to 1, with a fit index greater than .90 indicating good fit between the collected data and proposed model (Hu & Bentler, 1999). The amount of error variance in the model is reflected in the root mean square error of approximation (RMSEA); values less than .05 are considered adequate (Browne & Cudeck, 1993).

Results

Table 1 displays the Pearson product-moment correlations among the predictor variables along with the means and standard deviations for each variable. Independent samples *t* tests revealed there were no statistically significant differences for gender on any of the major study variables.

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Table 1. Correlations of emotional distress, cognitive appraisal, vision-related self-reported disability and visual acuity

Variable	1	2	3	4	5	*M*	*SD*
1. Emotional distress	–	−.44***	.46***	−.31**	.14	45.4	24.7
2. Tolerance		–	−.34***	.11	−.02	58.1	27.4
3. Interference			–	−.37***	.22*	69.7	28.9
4. Disability				–	−.53***	35.5	18.9
5. Visual acuity					–	1.03	0.38

Note. Lower scores on tolerance indicate greater difficulty in the ability to tolerate vision-related changes; higher scores for disability reflect fewer problems associated with vision-specific disability and higher scores for visual acuity indicate greater vision impairment.
*$p < .05$; **$p < .01$; ***$p < .001$.

To test our full theoretical model of the relationships among distress, appraisal, disability and vision symptom severity, we used maximum-likelihood estimates of the path model shown in Figure 1. The model parameter estimation was implemented with the AMOS (Version 5.0) statistical software package. In this model, the criterion variable was vision-related self-reported disability. The model assumes that the criterion is influenced by the direct effects of both emotional distress and objective visual acuity. The model also assumes that emotional distress is directly influenced by the appraisal variables (interference and tolerance). Based on the results of Harkins *et al.* (2006), we also included indirect effects of objective vision symptomatology (i.e. visual acuity) and tolerance on distress via the influence of these variables on appraisal of interference. Any missing values were imputed using maximum likelihood.

The model proved an acceptable fit to the data, $X^2(4) = 4.09$, $p = .39$. The CFI was .99 and the RMSEA was .01. Figure 1 presents the standardized parameter estimates for the paths of the model. All parameter estimates were statistically significant at least at the .01 α-level.

As indicated in Figure 1, both the direct effect of visual acuity ($-.50$, $p < .01$) and emotional distress ($-.22$, $p < .01$) on self-reported disability were statistically significant. Two-tailed significance tests for indirect effects were computed using bootstrapped standard errors (*SE*) for standardized effect estimates (based on 1,000 samples). All hypothesized indirect effects were statistically significant. The indirect effect of visual acuity on distress (via perceived interference) was .08 ($SE = .04$, $p < .05$), and its indirect effect on disability (via interference and emotional distress) was $-.02$ ($SE = .01$, $p < .05$). Thus the total effect of visual acuity on self-reported disability was $-.52$ ($SE = .08$, $p < .01$). Appraisals of how interfering vision loss was on everyday living had a significant indirect effect on disability (via emotional distress) of $-.08$ ($SE = .03$, $p < .01$), and the degree to which an individual was able to tolerate vision loss had a significant indirect effect on disability through interference and emotional distress of .10 ($SE = .03$, $p < .01$).

Modifications of this model were assessed for exploratory purposes, but none provided reasonable fit to the data. When visual acuity was assumed to have a direct rather than indirect effect on emotional distress, the standardized path coefficient was .01, *ns*, the CFI was .58, and the RMSEA was .25. This finding that objective symptomatology had no direct effect on emotional distress is consistent with the

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results of Harkins *et al.* (2006). When the direct path from visual acuity to vision-related self-reported disability was removed from the model, the CFI was .66 and the RMSEA .23, indicating that objective visual acuity has both direct and indirect effects on disability.

Discussion

These results provide evidence for the role of subjective appraisal mechanisms for understanding emotional distress and disability associated with vision loss. In general, our findings illustrate that cognitive appraisals of the degree to which individuals can tolerate vision loss and how interfering vision loss is on everyday living are major factors associated with the emotional distress. Distress subsequently predicts vision-related disability independent of actual impairment. Cognitive appraisals of interference and tolerance had significant indirect effects on disability. Severity of vision loss (e.g. acuity) was independently predictive of vision-related disability. Thus, severity of vision impairment does not necessarily correspond to emotional distress following vision loss. Rather, other important psychosocial factors such as subjective appraisals may influence emotional reactions to chronic health conditions and may provide a more informative account of a person's overall adjustment following vision loss.

From a theoretical standpoint, the findings support the component of the primary appraisal process detailed in the transactional model of stress appraisal and coping (Folkman & Moskowitz, 2004; Lazarus & Folkman, 1984). This activity is considered particularly important for understanding coping and adjustment in persons with chronic health conditions, generally (Elliott, Kurylo, & Rivera, 2002). While coping is relevant to understanding an individual's active strategies in managing stressors, knowledge of the individual meaning of adaptation provides insight into an individual's reactions to changing conditions and is not restricted to more active self-regulatory responses (Schilling & Whal, 2006). The current data suggest that those who perceive low vision as disruptive to expected behaviours, goals and outcomes might be expected to have recurring difficulties with distress. Emotional adjustment may be related to the extent to which one cognitively and or behaviourally modifies expectations for daily activities, functional abilities and desired outcomes (Williamson, 1998).

While the severity of vision loss was unrelated to the prediction of emotional distress, this finding has been observed in other studies. An explanation for the lack of relationship may be that many persons with severe physical problems do not necessarily perceive their physical problems as difficult to tolerate or interfering, in other words as the primary source of stress in their life, and those who do are more distressed emotionally in general (Elliott *et al.*, 2002). Thus, this supports the need for the understanding of the personal meaning a person attaches to vision loss as it relates to an individual's previously anticipated goals and expectancies is relevant.

It is also important to note that severity of vision loss was unrelated to tolerance appraisals. It may be that other factors are operating to influence this cognitive activity. For example, an individual with significantly impaired vision might not evaluate their situation involving vision loss as difficult to tolerate given the person has a strong social support network, adequate financial status and supportive work environment. In contrast, another person with the same degree of vision impairment might perceive their situation as greatly difficult to tolerate as they might not have the confidence or resources to adaptively cope.

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The current findings have several important implications for clinical work in low vision rehabilitation. For example, evaluating a person's appraisal of a chronic health condition (e.g. loss of vision) may be considered critical for identifying those who may be a risk for problems with emotional distress. Unfortunately, this type of assessment is not routinely conducted during low vision rehabilitation eye examinations. While distress may resolve in some persons without intervention, unmanaged and chronic distress may place a person at risk for depression and greater disability independent of vision loss. Thus, cognitive appraisal of interference and tolerance deserves attention during initial clinical eye examinations so that those who may be at risk are identified and referred for appropriate interventions. Administration of brief measures designed to evaluate appraisal processes and condition-specific distress such as the scales used in this study can be efficiently conducted in the clinic setting and quick to administer. Furthermore, the current findings should also be considered when designing rehabilitation programmes for individuals. Rehabilitation outcomes might be enhanced when rehabilitation programmes are tailored to the personal meaning and interpretation of vision loss on an individual's expected goals. Eliciting an individual's personal cognitive appraisals regarding vision loss might be an important initial step in low vision rehabilitation programmes prior to working on specific aspects of important rehabilitation services (i.e. occupational therapy, training with devices).

Previous research indicates that cognitive appraisals of stress (Shewchuk, Elliott, MacNair, & Harkins, 1999) and of physical symptoms (Harkins *et al.*, 2006) are rather stable over time, but there is evidence that these appraisals may be amenable to cognitive-behavioural interventions. For example, interventions such as problem-solving therapy (PST) to change self-appraised problem-solving abilities is an empirically supported intervention that may be useful in enhancing appraisal activity (Nezu, Felgoise, McClure, & Houts, 2003; Perri *et al.*, 2001). Other studies examining alternative treatment interventions indicate that individuals can be taught to accept difficult thoughts, feelings and bodily sensations without struggling with them, and focus on overt activities that contribute to important outcomes (i.e. Acceptance and Commitment Therapy; ACT) (Hayes, Strosahl, & Wilson, 1999). Therefore, it appears that some cognitive-behavioural interventions may offer some promise in working with personal appraisals of vision impairment. Generally, these approaches may be useful to: (1) minimize overall distress; (2) increase motivation for participation in rehabilitation efforts; (3) reevaluate goals and develop new goals and expectations; (4) identify alternative ways to tolerate and cope with stress and (5) increase self-efficacy in regulating emotions. These types of cognitive-behavioural interventions are particularly needed in visually impaired populations (Lueck, 1997).

In response to this need, several of the current authors are in the process of conducting a randomized clinical trial which aims to prevent distress by integrating PST along with occupational therapy. The treatment intervention is individually tailored to meet the unique problems and goals/appraisals of persons in low vision rehabilitation and is delivered by an occupational therapist who is part of the low vision rehabilitation treatment team. If successful, this type of service could help to enhance services to persons in low vision rehabilitation as occupational therapists are specifically trained to treat functional problems and work with the patient on developing obtainable goals. Thus, incorporating an emphasis on appraisal processes into OT services might help to augment treatment benefits.

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Several limitations related to this study exist. The sample was relatively homogeneous in age (primarily older adults), race (primarily Caucasian) and socio-economic status (able to afford low vision rehabilitation services). While this type of sample is representative for our centre given our current geographical location, future investigations are needed to determine whether these findings are generalizable to more demographically diverse samples. In addition, we captured persons seeking low vision rehabilitation services who may be distinctly different, in terms of appraisal processes, from those who do not seek services. Based on the literature demonstrating a relationship between appraisals and depression (i.e. Harkins *et al.*, 2006; Tolman *et al.*, 2005), one can speculate that appraisals may indeed impact motivation to seek low vision rehabilitation services. Further examination of appraisal processes in persons referred for low vision rehabilitation services who do not follow through and attend such services are warranted. Another limitation is related to the cross-sectional nature of the current study design. While a cross-sectional study can establish correlation and association, conclusions regarding causality cannot be determined. Prospective replications of the current findings and longitudinal studies would be beneficial. The study is limited to persons primarily who had central vision loss. Other factors that influence the person – environment dynamic of the transactional theory of coping should also be examined in conjunction with cognitive appraisals (i.e. personality, availability and type of services, social support).

While we attempted to expand upon previous research by using an 'objective' measure (i.e. visual acuity), investigations using other visual objective measures should also be further examined (i.e. contrast sensitivity). Investigations focusing on individuals with positive appraisals or resilient attitudes are also needed in order to develop interventions that promote successful adjustment.

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